

# NOTES

– NOTES WITH MIND MAPS –

# MATHEMATICS

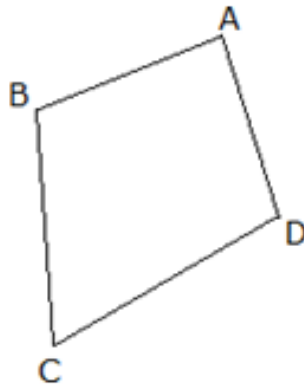
## (QUADRILATERALS)



## Quadrilaterals

### Quadrilateral

A **quadrilateral** is a closed figure obtained by joining four points (with no three points collinear) in an order.



Here, ABCD is a quadrilateral.

### Parts of a quadrilateral

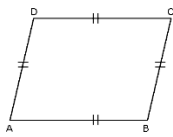
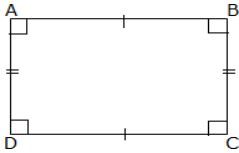
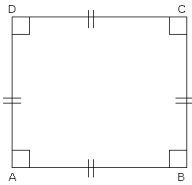
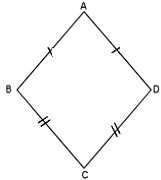
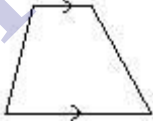
- A quadrilateral has four sides, four angles and four vertices.
- Two sides of a quadrilateral having no common end point are called its **opposite sides**.
- Two sides of a quadrilateral having a common end point are called its **adjacent sides**.
- Two angles of a quadrilateral having common arm are called its **adjacent angles**.
- Two angles of a quadrilateral not having a common arm are called its **opposite angles**.
- A **diagonal** is a line segment obtained on joining the opposite vertices.

### Angle sum property of a quadrilateral

Sum of all the angles of a quadrilateral is  $360^\circ$ . This is known as the **angle sum property of a quadrilateral**.

### Types of quadrilaterals and their properties:

| Name of a quadrilateral                                                       | Properties                                                                                           |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Parallelogram: A quadrilateral with each pair of opposite sides parallel.<br> | i. Opposite sides are equal.<br>ii. Opposite angles are equal.<br>iii. Diagonals bisect one another. |

|                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Rhombus:</b> A parallelogram with sides of equal length.</p>                                        | <ol style="list-style-type: none"> <li>All properties of a parallelogram.</li> <li>Diagonals are perpendicular to each other.</li> </ol>                                                                                                                     |
| <p><b>Rectangle:</b> A parallelogram with all angles right angle.</p>                                     | <ol style="list-style-type: none"> <li>All the properties of a parallelogram.</li> <li>Each of the angles is a right angle.</li> <li>Diagonals are equal.</li> </ol>                                                                                         |
| <p><b>Square:</b> A rectangle with sides of equal length.</p>                                             | <p>All the properties of a parallelogram, a rhombus and a rectangle.</p>                                                                                                                                                                                     |
| <p><b>Kite:</b> A quadrilateral with exactly two pairs of equal consecutive sides.</p>                  | <ol style="list-style-type: none"> <li>The diagonals are perpendicular to one another.</li> <li>One of the diagonals bisects the other.</li> <li>If ABCD is a kite, then <math>\angle B = \angle D</math> but <math>\angle A \neq \angle C</math></li> </ol> |
| <p><b>Trapezium:</b> A quadrilateral with one pair of opposite sides parallel is called trapezium.</p>  | <p>One pair of opposite sides parallel.</p>                                                                                                                                                                                                                  |

### Important facts about quadrilaterals

- If the non-parallel sides of trapezium are equal, it is known as **isosceles trapezium**.
- Square, rectangle and rhombus are all parallelograms.
- Kite and trapezium are not parallelograms.
- A square is a rectangle.
- A square is a rhombus.
- A parallelogram is a trapezium.

**A quadrilateral is a parallelogram if:**

- i. each pair of opposite sides of a quadrilateral is equal, or
- ii. each pair of opposite angles is equal, or
- iii. the diagonals of a quadrilateral bisect other, or
- iv. each pair of opposite sides is equal and parallel.

### Mid-Point Theorem

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and equal to half of it.

### Converse of mid-point theorem

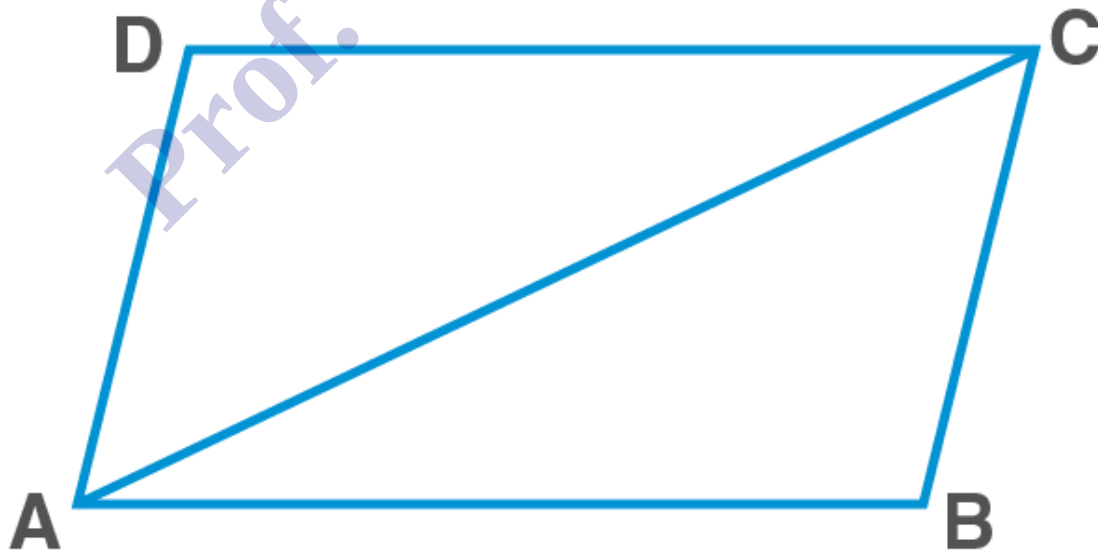
The line drawn through the mid-point of one side of a triangle, parallel to another side, bisects the third side.

### Formation of a new quadrilateral using the given data

- If the diagonals of a parallelogram are equal, then it is a rectangle.
- If the diagonals of a quadrilateral bisect each other at right angles, then it is a rhombus.
- If the diagonals of a quadrilateral are equal and bisect each other at right angles, then it is a square.

If there are three or more parallel lines and the intercepts made by them on a transversal are equal, then the corresponding intercepts on any other transversal are also equal.

### Parallelogram: Opposite sides of a parallelogram are equal



In  $\triangle ABC$  and  $\triangle CDA$

$AC = AC$  [Common / transversal]

$$\angle BCA = \angle DAC \text{ [alternate angles]}$$

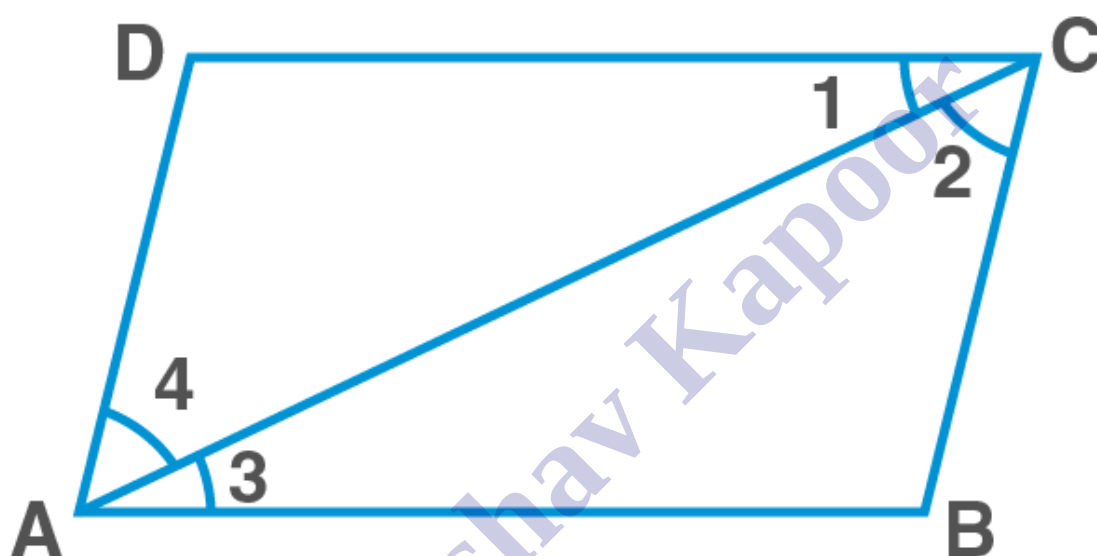
$$\angle BAC = \angle DCA \text{ [alternate angles]}$$

$$\triangle ABC \cong \triangle CDA \text{ [ASA rule]}$$

Hence,

$$AB = DC \text{ and } AD = BC \text{ [ C.P.C.T.C]}$$

Opposite angles in a parallelogram are equal



In parallelogram ABCD

$AB \parallel CD$ ; and AC is the transversal

Hence,  $\angle 1 = \angle 3$  .... (1) (alternate interior angles)

$BC \parallel DA$ ; and AC is the transversal

Hence,  $\angle 2 = \angle 4$  .... (2) (alternate interior angles)

Adding (1) and (2)

$$\angle 1 + \angle 2 = \angle 3 + \angle 4$$

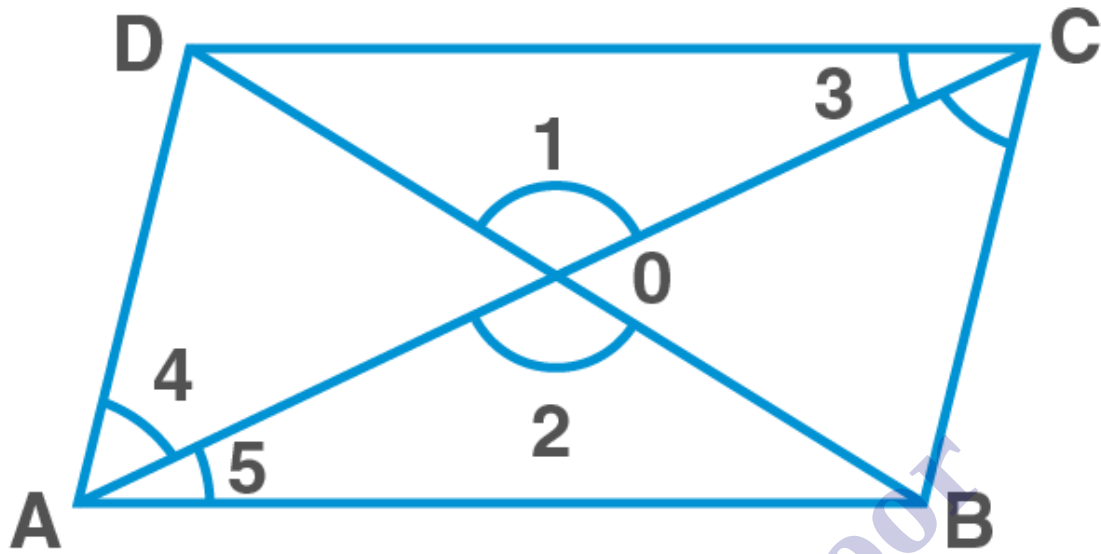
$$\angle BAD = \angle BCD$$

Similarly,

$$\angle ADC = \angle ABC$$

**Properties of diagonal of a parallelogram**

Diagonals of a parallelogram bisect each other.



In  $\triangle AOB$  and  $\triangle COD$ ,

$\angle 3 = \angle 5$  [alternate interior angles]

$\angle 1 = \angle 2$  [vertically opposite angles]

$AB = CD$  [opp. Sides of parallelogram]

$\triangle AOB \cong \triangle COD$  [AAS rule]

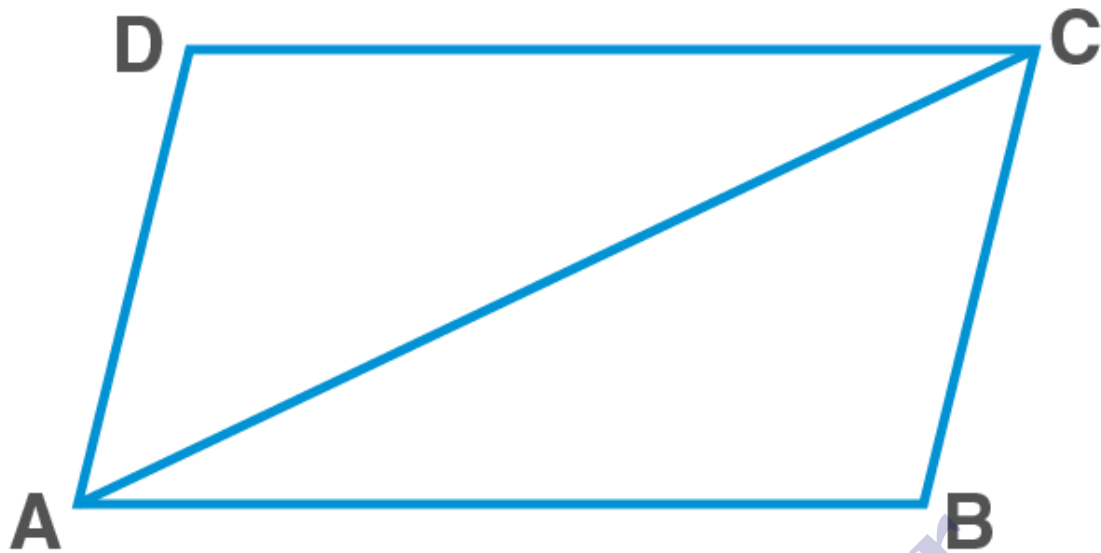
$OB = OD$  and  $OA = OC$  [C.P.C.T]

Hence, proved

Conversely,

If the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Diagonal of a parallelogram divides it into two congruent triangles.



In  $\triangle ABC$  and  $\triangle CDA$ ,

$AB = CD$  [Opposite sides of parallelogram]

$BC = AD$  [Opposite sides of parallelogram]

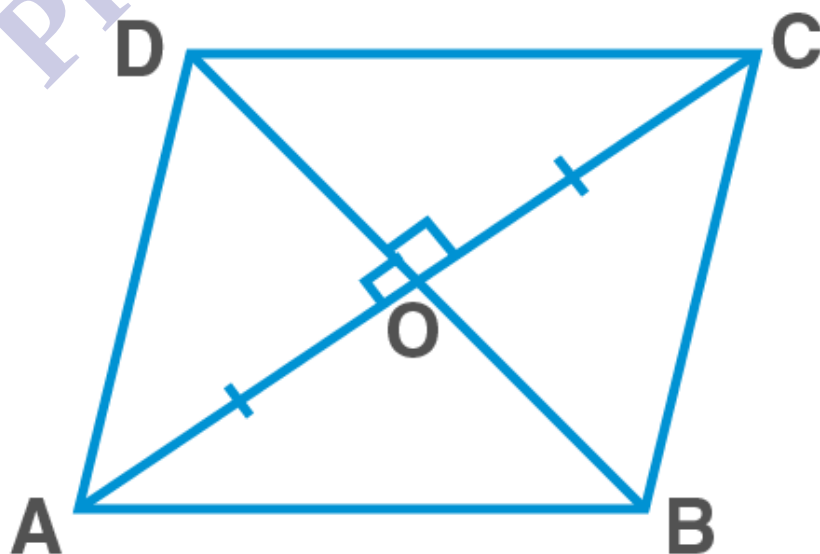
$AC = AC$  [Common side]

$\triangle ABC \cong \triangle CDA$  [by SSS rule]

Hence, proved.

**Diagonals of a rhombus bisect each other at right angles**

Diagonals of a rhombus bisect each – other at right angles



In  $\triangle AOD$  and  $\triangle COD$ ,

$OA = OC$  [Diagonals of parallelogram bisect each other]

$OD = OD$  [Common side]

$AD = CD$  [Adjacent sides of a rhombus]

$\triangle AOD \cong \triangle COD$  [SSS rule]

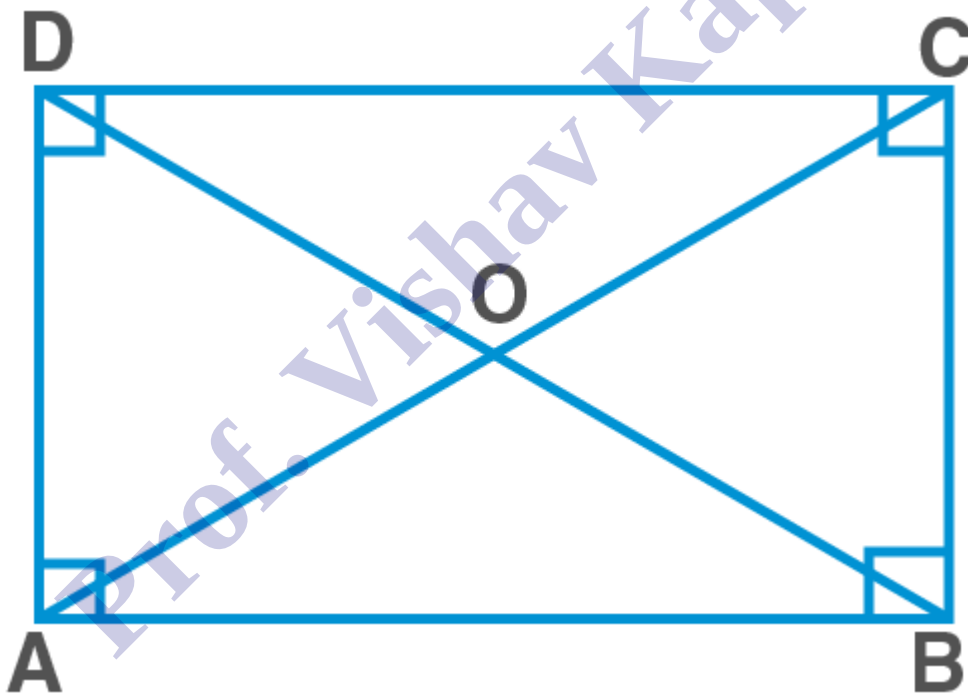
$\angle AOD = \angle DOC$  [C.P.C.T]

$\angle AOD + \angle DOC = 180$  [ $\because$  AOC is a straight line]

Hence,  $\angle AOD = \angle DOC = 90$

Hence proved.

**Diagonals of a rectangle bisect each other and are equal**



Rectangle ABCD

In  $\triangle ABC$  and  $\triangle BAD$ ,

$AB = BA$  [Common side]

$BC = AD$  [Opposite sides of a rectangle]

$\angle ABC = \angle BAD$  [Each =  $90^\circ \because$  ABCD is a Rectangle]

$\triangle ABC \cong \triangle BAD$  [SAS rule]



$$\therefore AC = BD \text{ [C.P.C.T]}$$

Consider  $\triangle OAD$  and  $\triangle OCB$ ,

$$AD = CB \text{ [Opposite sides of a rectangle]}$$

$$\angle OAD = \angle OCB \text{ [}\because AD \parallel BC \text{ and transversal AC intersects them]}$$

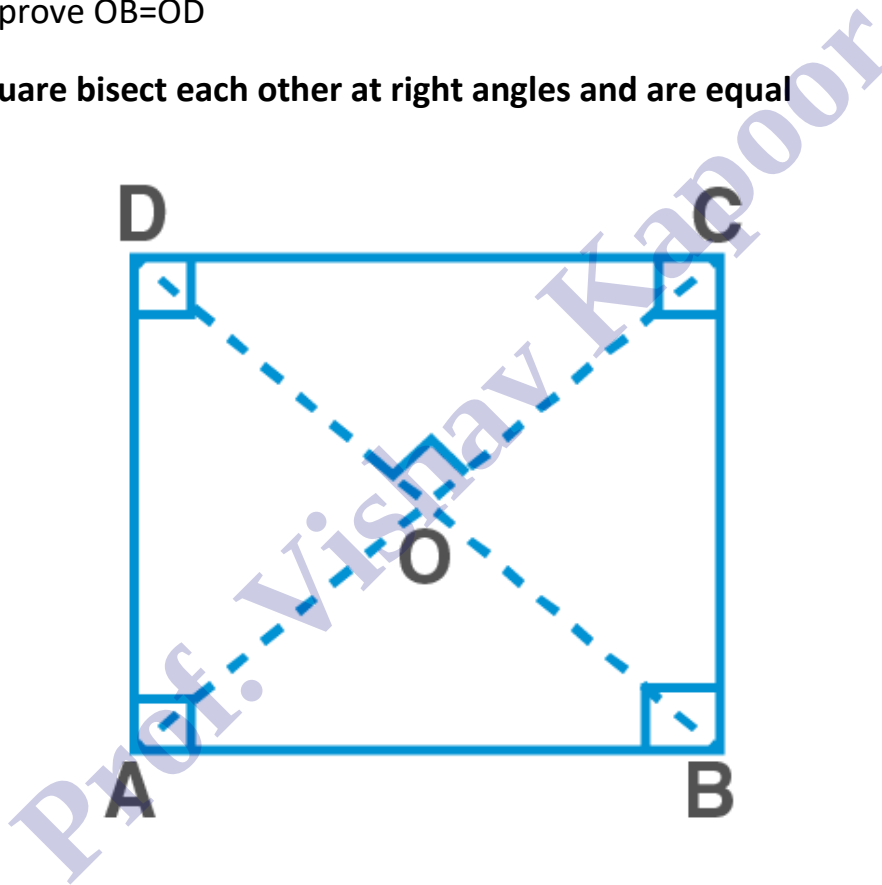
$$\angle ODA = \angle OBC \text{ [}\because AD \parallel BC \text{ and transversal BD intersects them]}$$

$$\triangle OAD \cong \triangle OCB \text{ [ASA rule]}$$

$$\therefore OA = OC \text{ [C.P.C.T]}$$

Similarly, we can prove  $OB = OD$

**Diagonals of a square bisect each other at right angles and are equal**



In  $\triangle ABC$  and  $\triangle BAD$ ,

$$AB = BA \text{ [Common side]}$$

$$BC = AD \text{ [Opposite sides of a Square]}$$

$$\angle ABC = \angle BAD \text{ [Each} = 90^\circ \because ABCD \text{ is a Square]}$$

$$\triangle ABC \cong \triangle BAD \text{ [SAS rule]}$$

$$\therefore AC = BD \text{ [C.P.C.T]}$$

Consider  $\triangle OAD$  and  $\triangle OCB$ ,

$AD = CB$  [Opposite sides of a Square]

$\angle OAD = \angle OCB$  [ $\because AD \parallel BC$  and transversal  $AC$  intersects them]

$\angle ODA = \angle OBC$  [ $\because AD \parallel BC$  and transversal  $BD$  intersects them]

$\triangle OAD \cong \triangle OCB$  [ASA rule]

$\therefore OA = OC$  [C.P.C.T]

Similarly, we can prove  $OB = OD$

In  $\triangle OBA$  and  $\triangle ODA$ ,

$OB = OD$  [proved above]

$BA = DA$  [Sides of a Square]

$OA = OA$  [Common side]

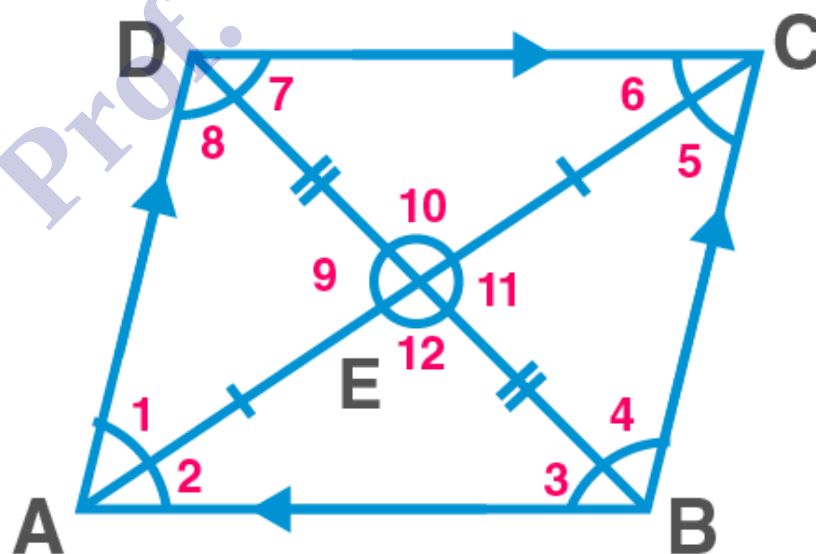
$\triangle OBA \cong \triangle ODA$ , [SSS rule]

$\therefore \angle AOB = \angle AOD$  [C.P.C.T]

But  $\angle AOB + \angle AOD = 180^\circ$  [Linear pair]

$\therefore \angle AOB = \angle AOD = 90^\circ$

Important results related to parallelograms



Opposite sides of a parallelogram are parallel and equal.

$AB \parallel CD, AD \parallel BC, AB = CD, AD = BC$

Opposite angles of a parallelogram are equal adjacent angles are supplementary.

$$\angle A = \angle C, \angle B = \angle D,$$

$$\angle A + \angle B = 180^\circ, \angle B + \angle C = 180^\circ, \angle C + \angle D = 180^\circ, \angle D + \angle A = 180^\circ$$

A diagonal of parallelogram divides it into two congruent triangles.

$$\triangle ABC \cong \triangle CDA \text{ [With respect to AC as diagonal]}$$

$$\triangle ADB \cong \triangle CBD \text{ [With respect to BD as diagonal]}$$

The diagonals of a parallelogram bisect each other.

$$AE = CE, BE = DE$$

$$\angle 1 = \angle 5 \text{ (alternate interior angles)}$$

$$\angle 2 = \angle 6 \text{ (alternate interior angles)}$$

$$\angle 3 = \angle 7 \text{ (alternate interior angles)}$$

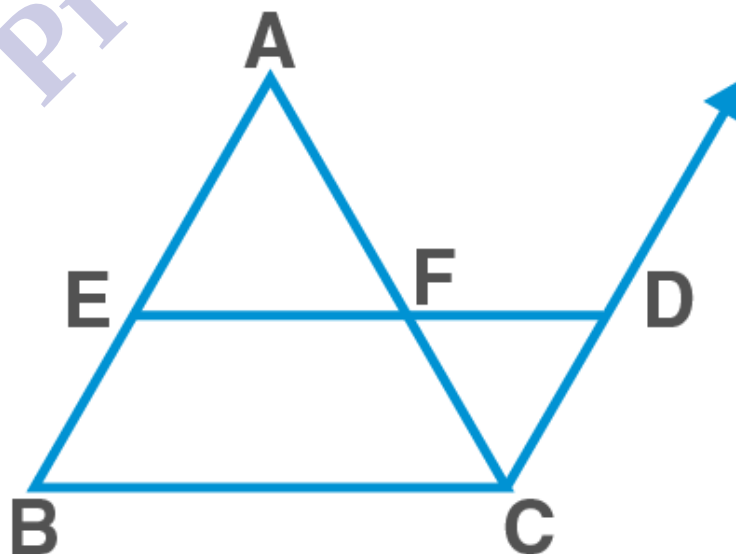
$$\angle 4 = \angle 8 \text{ (alternate interior angles)}$$

$$\angle 9 = \angle 11 \text{ (vertically opp. angles)}$$

$$\angle 10 = \angle 12 \text{ (vertically opp. angles)}$$

### The Mid-Point Theorem

The line segment joining the midpoints of two sides of a triangle is parallel to the third side and is half of the third side



In  $\triangle ABC$ , E – the midpoint of AB; F – the midpoint of AC

Construction: Produce EF to D such that EF = DF.

In  $\triangle AEF$  and  $\triangle CDF$ ,

AF = CF [F is the midpoint of AC]

$\angle AFE = \angle CFD$  [V.O.A]

EF = DF [Construction]

$\therefore \triangle AEF \cong \triangle CDF$  [SAS rule]

Hence,

$\angle EAF = \angle DCF$  .... (1)

DC = EA = EB [E is the midpoint of AB]

DC  $\parallel$  EA  $\parallel$  AB [Since, (1), alternate interior angles]

DC  $\parallel$  EB

So EBCD is a parallelogram

Therefore, BC = ED and BC  $\parallel$  ED

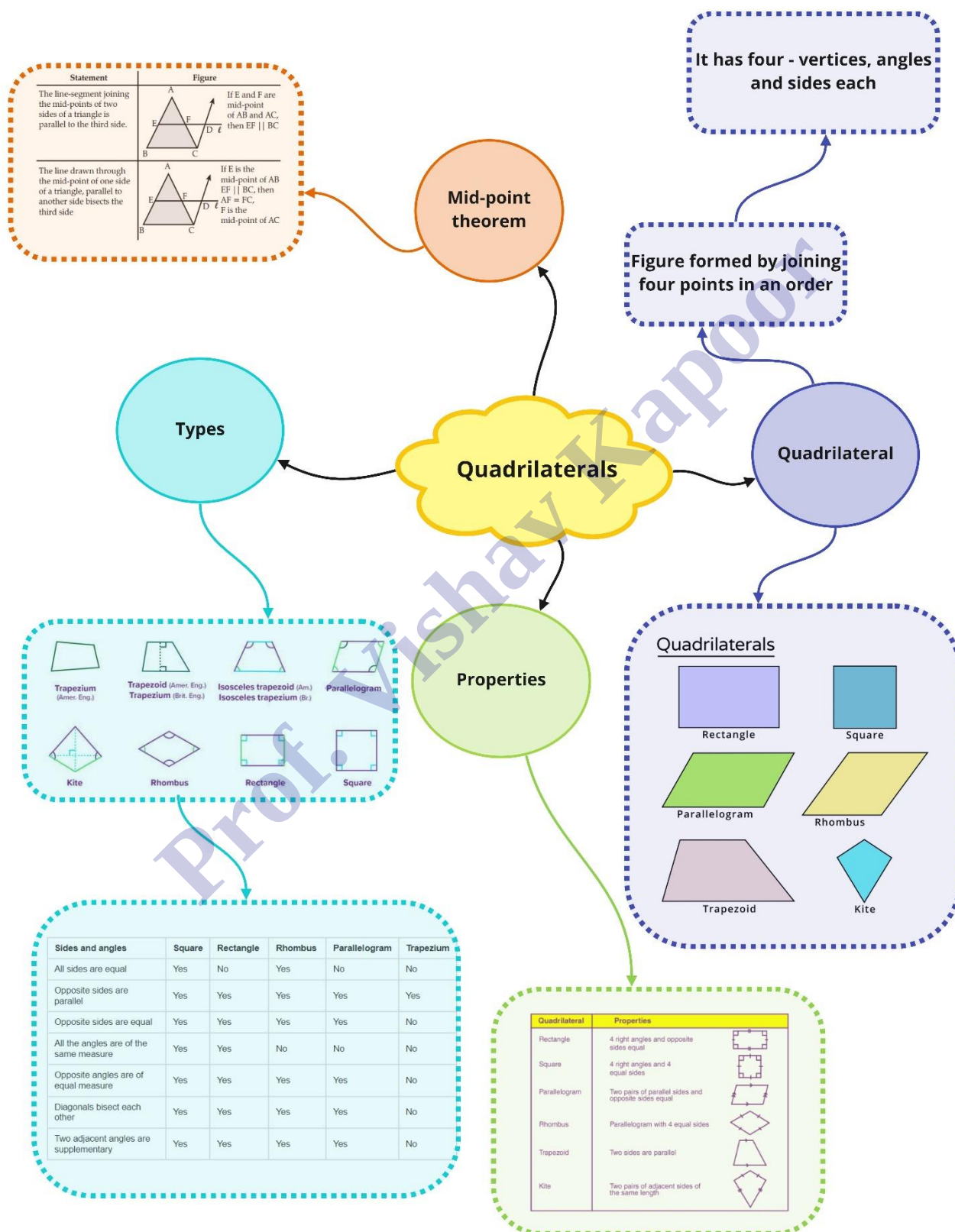
Since ED = EF + FD = 2EF = BC [  $\because$  EF=FD]

We have, EF =  $\frac{1}{2}$  BC and EF  $\parallel$  BC

Hence proved.

Prof. Vishav Kapoor

Class : 9th mathematics  
Chapter- 8: Quadrilaterals



## Important Questions

### Multiple Choice questions-

Question 1. A diagonal of a Rectangle is inclined to one side of the rectangle at an angle of  $25^\circ$ . The Acute Angle between the diagonals is:

- (a)  $115^\circ$
- (b)  $50^\circ$
- (c)  $40^\circ$
- (d)  $25^\circ$

Question 2. The diagonals of a rectangle PQRS intersect at O. If  $\angle QOR = 44^\circ$ ,  $\angle OPS = ?$

- (a)  $82^\circ$
- (b)  $52^\circ$
- (c)  $68^\circ$
- (d)  $75^\circ$

Question 3. If angles A, B, C and D of the quadrilateral ABCD, taken in order, are in the ratio 3:7:6:4, then ABCD is

- (a) Rhombus
- (b) Parallelogram
- (c) Trapezium
- (d) Kite

Question 4. All the angles of a convex quadrilateral are congruent. However, not all its sides are congruent. What type of quadrilateral is it?

- (a) Parallelogram
- (b) Square
- (c) Rectangle
- (d) Trapezium

Question 5. In a Quadrilateral ABCD,  $AB = BC$  and  $CD = DA$ , then the quadrilateral is a

- (a) Triangle
- (b) Kite
- (c) Rhombus
- (d) Rectangle

Question 6. The angles of a quadrilateral are  $(5x)^\circ$ ,  $(3x + 10)^\circ$ ,  $(6x - 20)^\circ$  and  $(x$

+ 25)°. Now, the measure of each angle of the quadrilateral will be

- (a) 115°, 79°, 118°, 48°
- (b) 100°, 79°, 118°, 63°
- (c) 110°, 84°, 106°, 60°
- (d) 75°, 89°, 128°, 68°

Question 7. The diagonals of rhombus are 12 cm and 16 cm. The length of the side of rhombus is:

- (a) 12cm
- (b) 16cm
- (c) 8cm
- (d) 10cm

Question 8. In quadrilateral PQRS, if  $\angle P = 60^\circ$  and  $\angle Q : \angle R : \angle S = 2 : 3 : 7$ , then  $\angle S =$

- (a) 175°
- (b) 210°
- (c) 150°
- (d) 135°

Question 9. In parallelogram ABCD, if  $\angle A = 2x + 15^\circ$ ,  $\angle B = 3x - 25^\circ$ , then value of x is:

- (a) 91°
- (b) 89°
- (c) 34°
- (d) 38°

Question 10. If ABCD is a trapezium in which  $AB \parallel CD$  and  $AD = BC$ , then:

- (a)  $\angle A = \angle B$
- (b)  $\angle A > \angle B$
- (c)  $\angle A < \angle B$
- (d) None of the above

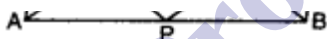
### Very Short:

1. If one angle of a parallelogram is twice of its adjacent angle, find the angles of the parallelogram.
2. If the diagonals of a quadrilateral bisect each other at right angles, then name the quadrilateral.

- Three angles of a quadrilateral are equal, and the fourth angle is equal to  $144^\circ$ . Find each of the equal angles of the quadrilateral.
- If ABCD is a parallelogram, then what is the measure of  $\angle A - \angle C$ ?
- PQRS is a parallelogram, in which  $PQ = 12$  cm and its perimeter is 40 cm. Find the length of each side of the parallelogram.
- Two consecutive angles of a parallelogram are  $(x + 60)^\circ$  and  $(2x + 30)^\circ$ . What special name can you give to this parallelogram?
- ONKA is a square with  $\angle KON = 45^\circ$ . Determine  $\angle KOA$ .
- In quadrilateral PQRS, if  $\angle P = 60^\circ$  and  $\angle Q : \angle R : \angle S = 2 : 3 : 7$ , then find the measure of  $\angle S$ .

### Short Questions:

- ABCD is a parallelogram in which  $\angle ADC = 75^\circ$  and side AB is produced to point E as shown in the figure. Find  $x + y$ .
- If the diagonals of a parallelogram are equal, then show that it is a rectangle.
- In the figure, ABCD is a rhombus, whose diagonals meet at O. Find the values of  $x$  and  $y$ .
- ABCD is a parallelogram and AP and CQ are perpendiculars from vertices A and C on diagonal BD (see fig.). Show that:
  - $\triangle APB \cong \triangle CQD$
  - $AP = CQ$
- The diagonals of a quadrilateral ABCD are perpendicular to each other. Show that the quadrilateral formed by joining the mid-points of its sides is a rectangle.



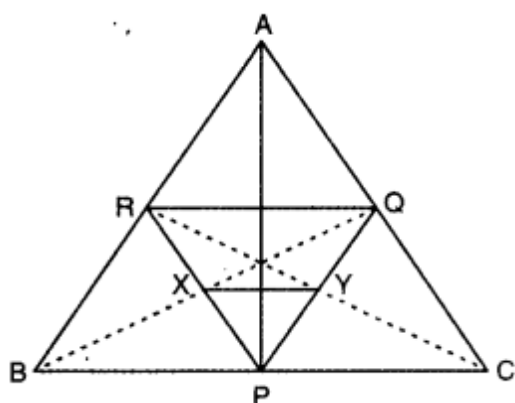
- In the fig., D, E and F are, respectively the mid-points of sides BC, CA and AB of an equilateral triangle ABC. Prove that DEF is also an equilateral triangle.



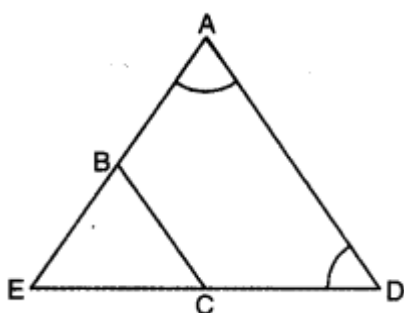
### Long Questions:

- In the figure, P, Q and R are the mid-points of the sides BC, AC and AB of  $\triangle ABC$ . If BQ and PR intersect at X and CR and PQ intersect at Y, then show that  $XY = \frac{1}{4} BC$

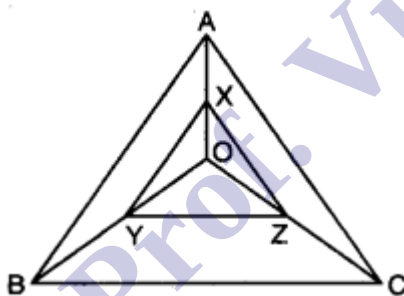




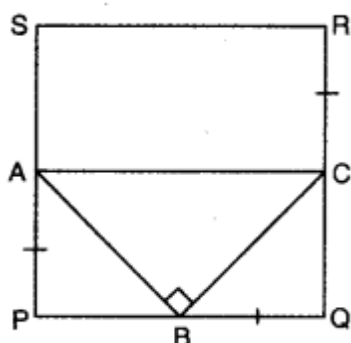
2. In the given figure,  $AE = DE$  and  $BC \parallel AD$ . Prove that the points A, B, C and D are concyclic. Also, prove that the diagonals of the quadrilateral ABCD are equal.



3. In  $\triangle ABC$ ,  $AB = 8\text{cm}$ ,  $BC = 9\text{cm}$  and  $AC = 10\text{cm}$ . X, Y and Z are mid-points of AO, BO and CO respectively as shown in the figure. Find the lengths of the sides of  $\triangle XYZ$ .



4. PQRS is a square and  $\angle ABC = 90^\circ$  as shown in the figure. If  $AP = BQ = CR$ , then prove that  $\angle BAC = 45^\circ$



5. ABCD is a parallelogram. If the bisectors DP and CP of angles D and C meet at P on side AB, then show that P is the mid-point of side AB.

## Assertion and Reason Questions-

1. In these questions, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- c) Assertion is correct statement but reason is wrong statement.
- d) Assertion is wrong statement but reason is correct statement.

**Assertion:** ABCD is a square. AC and BD intersect at O. The measure of  $\angle AOB = 90^\circ$ .

**Reason:** Diagonals of a square bisect each other at right angles.

2. In these questions, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- c) Assertion is correct statement but reason is wrong statement.
- d) Assertion is wrong statement but reason is correct statement.

**Assertion:** The consecutive sides of a quadrilateral have one common point.

**Reason:** The opposite sides of a quadrilateral have two common point.

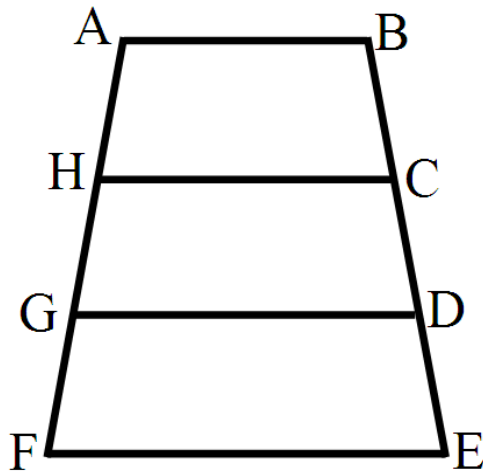
## Case Study Questions-

1. Read the Source/ Text given below and answer these questions:



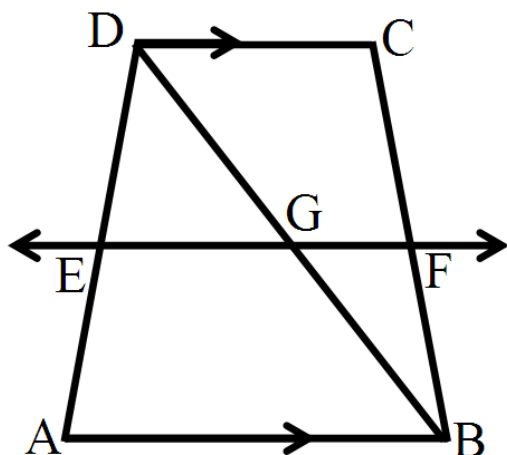
Sohan wants to show gratitude towards his teacher by giving her a card made by him. He has three pieces of trapezium pasted one above the other as shown in fig. These pieces are arranged in a way that  $AB \parallel HC \parallel GD \parallel FE$ . Also  $BC =$

CD = DE and AH = HG = GF = 6cm. He wants to decorate the card by putting up a colored tape on the nonparallel sides of the trapezium.



- i. Find the total length of colored tape required if  $DE = 4\text{cm}$ .
  - a. 20cm
  - b. 30cm
  - c. 40cm
  - d. 50cm
- ii. ABHC is a trapezium in which  $AB \parallel HC$  and  $\angle A = \angle B = 45^\circ$ . Find angles C and H of the trapezium.
  - a. 135, 130
  - b. 130, 135
  - c. 135, 135
  - d. 130, 130
- iii. What is the difference between trapezium and parallelogram?
  - a. Trapezium has 2 sides, and parallelogram has 4 sides.
  - b. Trapezium has 4 sides, and parallelogram has 2 sides.
  - c. Trapezium has 1 pair of parallel sides, and parallelogram has 2 pairs of parallel sides.
  - d. Trapezium has 2 pairs of parallel sides, and parallelogram has 1 pair of parallel sides.
- iv. Diagonals in isosceles trapezoid are \_\_\_\_\_.
  - a. parallel.
  - b. opposite.
  - c. vertical.
  - d. equal.

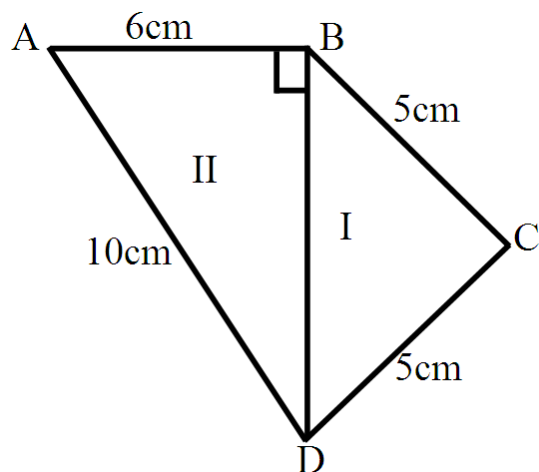
- v. ABCD is a trapezium where  $AB \parallel DC$ , BD is the diagonal and E is the midpoint of AD. A line is drawn through E parallel to AB intersecting BC at F. Which of these is true?



- a.  $BF = FC$
  - b.  $EA = FB$
  - c.  $CF = DE$
  - d. None of these
2. Read the Source/ Text given below and answer any four questions:



Chocolate is in the form of a quadrilateral with sides 6cm and 10cm, 5cm and 5cm(as shown in the figure) is cut into two parts on one of its diagonal by a lady. Part-I is given to her maid and part II is equally divided among a driver and gardener.



- i. Length of BD:
  - a. 9cm
  - b. 8cm
  - c. 7cm
  - d. 6cm
- ii. Area of  $\triangle ABC$ :
  - a.  $24\text{cm}^2$
  - b.  $12\text{cm}^2$
  - c.  $42\text{cm}^2$
  - d.  $21\text{cm}^2$
- iii. The sum of all the angles of a quadrilateral is equal to:
  - a.  $180^\circ$
  - b.  $270^\circ$
  - c.  $360^\circ$
  - d.  $90^\circ$
- iv. A diagonal of a parallelogram divides it into two congruent:
  - a. Square.
  - b. Parallelogram.
  - c. Triangles.
  - d. Rectangle.
- v. Each angle of the rectangle is:
  - a. More than  $90^\circ$
  - b. Less than  $90^\circ$
  - c. Equal to  $90^\circ$

d. Equal to  $45^\circ$ **Answer Key:****MCQ:**

1. (b)  $50^\circ$
2. (c)  $68^\circ$
3. (c) Trapezium
4. (c) Rectangle
5. (b) Kite
6. (a)  $115^\circ, 79^\circ, 118^\circ, 48^\circ$
7. (d) 10cm
8. (a)  $175^\circ$
9. (d)  $38^\circ$
10. (a)  $\angle A = \angle B$

**Very Short Answer:**

1. Let the two adjacent angles be  $x$  and  $2x$ .

In a parallelogram, sum of the adjacent angles are  $180^\circ$

$$\therefore x + 2x = 180^\circ$$

$$\Rightarrow 3x = 180^\circ$$

$$\Rightarrow x = 60^\circ$$

Thus, the two adjacent angles are  $120^\circ$  and  $60^\circ$ . Hence, the angles of the parallelogram are  $120^\circ, 60^\circ, 120^\circ$  and  $60^\circ$ .

2. Rhombus.

3. Let each equal angle of given quadrilateral be  $x$ .

We know that sum of all interior angles of a quadrilateral is  $360^\circ$

$$\therefore x + x + x + 144^\circ = 360^\circ$$

$$3x = 360^\circ - 144^\circ$$

$$3x = 216^\circ$$

$$x = 72^\circ$$

Hence, each equal angle of the quadrilateral is of  $72^\circ$  measures.

4.  $\angle A - \angle C = 0^\circ$  (opposite angles of parallelogram are equal)

- 5.

$$12 \text{ cm}$$

Here,  $PQ = SR = 12 \text{ cm}$

Let  $PS = x$  and  $PS = QR$

$$\therefore x + 12 + x + 12 = \text{Perimeter}$$

$$2x + 24 = 40$$

$$2x = 16$$

$$x = 8$$

Hence, length of each side of the parallelogram is 12cm, 8 cm, 12cm and 8cm.

6. We know that consecutive interior angles of a parallelogram are supplementary.

$$\therefore (x + 60^\circ) + (2x + 30^\circ) = 180^\circ$$

$$\Rightarrow 3x^\circ + 90^\circ = 180^\circ$$

$$\Rightarrow 3x^\circ = 90^\circ$$

$$\Rightarrow x^\circ = 30^\circ$$

Thus, two consecutive angles are  $(30 + 60)^\circ$ ,  $12 \times 30 + 30)^\circ$ . i.e.,  $90^\circ$  and  $90^\circ$ .

Hence, the special name of the given parallelogram is rectangle.

7. Since ONKA is a square

$$\therefore \angle AON = 90^\circ$$

We know that diagonal of a square bisects its  $\angle$ s

$$\Rightarrow \angle AOK = \angle KON = 45^\circ$$

$$\text{Hence, } \angle KOA = 45^\circ$$

$$\text{Now, } \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle A + 70^\circ + 70^\circ = 180^\circ$$

$$[\because \angle B = 70^\circ]$$

$$\Rightarrow \angle A = 180^\circ - 70^\circ - 70^\circ = 40^\circ$$

8. Let  $\angle Q = 2x$ ,  $\angle R = 3x$  and  $\angle S = 7x$

$$\text{Now, } \angle P + \angle Q + \angle R + \angle S = 360^\circ$$

$$\Rightarrow 60^\circ + 2x + 3x + 7x = 360^\circ$$

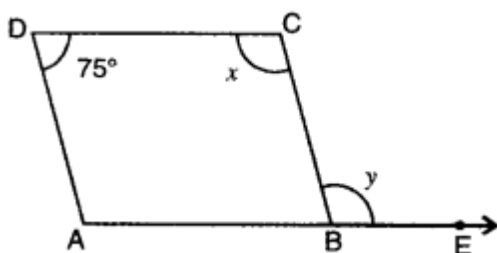
$$\Rightarrow 12x = 300^\circ$$

$$x = \frac{300^\circ}{12} = 25^\circ$$

$$\angle S = 7x = 7 \times 25^\circ = 175^\circ$$

### Short Answer:

Ans: 1.



Here,  $\angle C$  and  $\angle D$  are adjacent angles of the parallelogram.

$$\therefore \angle C + \angle D = 180^\circ$$

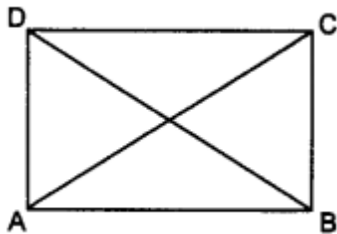
$$\Rightarrow x + 75^\circ = 180^\circ$$

$$\Rightarrow x = 105^\circ$$

$$\text{Also, } y = x = 105^\circ \text{ [alt. int. angles]}$$

$$\text{Thus, } x + y = 105^\circ + 105^\circ = 210^\circ$$

**Ans: 2.**



Given: A parallelogram ABCD, in which  $AC = BD$ .

To Prove:  $\triangle BCD$  is a rectangle.

Proof: In  $\triangle ABC$  and  $\triangle BAD$

$$AB = AB \text{ (common)}$$

$$AC = BD \text{ (given)}$$

$$BC = AD \text{ (opp. sides of a || gm)}$$

$$\Rightarrow \triangle ABC \cong \triangle BAD$$

[by SSS congruence axiom]

$$\Rightarrow \angle ABC = \angle BAD \text{ (c.p.c.t.)}$$

$$\text{Also, } \angle ABC + \angle BAD = 180^\circ \text{ (co-interior angles)}$$

$$\angle ABC + \angle ABC = 180^\circ \text{ [} \because \angle ABC = \angle BAD \text{]}$$

$$2\angle ABC = 180^\circ$$

$$\angle ABC = \frac{1}{2} \times 180^\circ = 90^\circ$$

Hence, parallelogram ABCD is a rectangle.

**Ans: 3.** Since diagonals of a rhombus bisect each other at right angle.

In  $\therefore \triangle AOB$ , we have

$$\angle OAB + \angle x + 90^\circ = 180^\circ$$

$$\angle x = 180^\circ - 90^\circ - 35^\circ$$

$$= 55^\circ$$

Also,

$$\angle DAO = \angle BAO = 35^\circ$$

$$\angle y + \angle DAO + \angle BAO + \angle x = 180^\circ$$

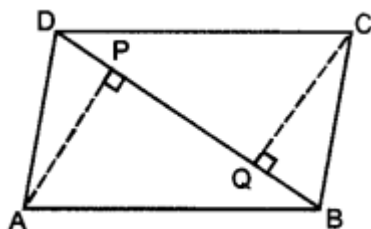


$$\Rightarrow \angle y + 35^\circ + 35^\circ + 55^\circ = 180^\circ$$

$$\Rightarrow \angle y = 180^\circ - 125^\circ = 55^\circ$$

Hence, the values of x and y are  $x = 55^\circ$ ,  $y = 55^\circ$

**Ans: 4.**



Given: In  $\parallel\text{gm}$  ABCD, AP and CQ are perpendiculars from the vertices A and C on the diagonal BD.

To Prove: (i)  $\triangle APB \cong \triangle CQD$

(ii)  $AP = CQ$

Proof: (i) In  $\triangle APB$  and  $\triangle CQD$

$AB = DC$  (opp. sides of a  $\parallel\text{gm}$  ABCD]

$\angle APB = \angle DQC$  (each  $= 90^\circ$ )

$\angle ABP = \angle CDQ$  (alt. int.  $\angle$ s]

$\Rightarrow \triangle APB \cong \triangle CQD$  [by AAS congruence axiom]

(ii)  $\Rightarrow AP = CQ$  [c.p.c.t.]

**Ans: 5.** Given: A quadrilateral ABCD whose diagonals AC and BD are perpendicular to each other at O. P, Q, R and S are mid-points of side AB, BC, CD and DA respectively are joined are formed quadrilateral PQRS.

To Prove: PQRS is a rectangle.

Proof: In  $\triangle ABC$ , P and Q are mid-points of AB and BC respectively.

$\therefore PQ \parallel AC$  and  $PQ = \frac{1}{2} AC$  ... (i) (mid-point theorem]

Further, in  $\triangle ACD$ , R and S are mid-points of CD and DA respectively.

$SR \parallel AC$  and  $SR = \frac{1}{2} AC$  ... (ii) (mid-point theorem]

From (i) and (ii), we have  $PQ \parallel SR$  and  $PQ = SR$

Thus, one pair of opposite sides of quadrilateral PQRS are parallel and equal.

$\therefore$  PQRS is a parallelogram.

Since  $PQ \parallel AC$  and  $SR \parallel AC$

In  $\triangle ABD$ , P and S are mid-points of AB and AD respectively.

$PS \parallel BD$  (mid-point theorem]

$$\Rightarrow PN \parallel MO$$

$\therefore$  Opposite sides of quadrilateral PMON are parallel.

$\therefore$  PMON is a parallelogram.

$$\angle MPN = \angle MON \text{ (opposite angles of } \parallel \text{ gm are equal)}$$

$$\text{But } \angle MON = 90^\circ \text{ [given]}$$

$$\therefore \angle MPN = 90^\circ \Rightarrow \angle QPS = 90^\circ$$

Thus, PQRS is a parallelogram whose one angle is  $90^\circ$

$\therefore$  PQRS is a rectangle.

**Ans: 6.** Since line segment joining the mid-points of two sides of a triangle is half of the third side.

Therefore, D and E are mid-points of BC and AC respectively.

$$\Rightarrow DE = \frac{1}{2} AB \dots (i)$$

E and F are the mid-points of AC and AB respectively.

$$\therefore EF = \frac{1}{2} BC \dots (ii)$$

F and D are the mid-points of AB and BC respectively.

$$\therefore FD = \frac{1}{2} AC \dots (iii)$$

Now,  $\triangle ABC$  is an equilateral triangle.

$$\Rightarrow AB = BC = CA$$

$$\Rightarrow \frac{1}{2} AB = \frac{1}{2} BC = \frac{1}{2} CA$$

$$\Rightarrow DE = EF = FD \text{ (using (i), (ii) and (iii))}$$

Hence, DEF is an equilateral triangle

### Long Answer:

**Ans: 1.** Here, in  $\triangle ABC$ , R and Q are the mid-points of AB and AC respectively.

$\therefore$  By using mid-point theorem, we have

$$RQ \parallel BC \text{ and } RQ = \frac{1}{2} BC$$

$$\therefore RQ = BP = PC \text{ [}\because \text{ P is the mid-point of BC]}$$

$$\therefore RQ \parallel BP \text{ and } RQ \parallel PC$$

In quadrilateral BPQR

$$RQ \parallel BP, RQ = BP \text{ (proved above)}$$

$\therefore$  BPQR is a parallelogram. [ $\because$  one pair of opp. sides is parallel as well as equal]

$\therefore$  X is the mid-point of PR. [ $\because$  diagonals of a  $\parallel$  gm bisect each other]

Now, in quadrilateral PCQR

$RQ \parallel PC$  and  $RQ = PC$  [proved above]

$\therefore$  PCQR is a parallelogram [ $\because$  one pair of opp. sides is parallel as well as equal]

$\therefore$  Y is the mid-point of PQ [ $\because$  diagonals of a ||gm bisect each other]

In  $\Delta PQR$

$\therefore$  X and Y are mid-points of PR and PQ respectively.

$\therefore XY \parallel RQ$  and  $XY = \frac{1}{2} RQ$  [by using mid-point theorem]

$$XY = \frac{1}{2} \left( \frac{1}{2} BC \right) \quad [\because RQ = \frac{1}{2} BC]$$

$$\Rightarrow XY = \frac{1}{4} BC$$

**Ans: 2.** Since  $AE = DE$

$\angle D = \angle A$  .... (i) [ $\because$   $\angle$ s opp. to equal sides of a  $\Delta$ ]

Again,  $BC \parallel AD$

$\angle EBC = \angle A$  .... (ii) (corresponding  $\angle$ s)

From (i) and (ii), we have

$\angle D = \angle EBC$  .... (iii)

But  $\angle EBC + \angle ABC = 180^\circ$  (a linear pair)

$\angle D + \angle ABC = 180^\circ$  (using (iii))

Now, a pair of opposite angles of quadrilateral ABCD is supplementary

Thus, ABCD is a cyclic quadrilateral i.e., A, B, C and D are concyclic. In  $\Delta ABD$  and  $\Delta DCA$

$\angle ABD = \angle ACD$  [ $\angle$ s in the same segment for cyclic quad. ABCD]

$\angle BAD = \angle CDA$  [using (i)]

$AD = AD$  (common)

So, by using AAS congruence axiom, we have

$\Delta ABD \cong \Delta DCA$

Hence,  $BD = CA$  [c.p.c.t.]

**Ans: 3.** Here, in  $\Delta ABC$ ,  $AB = 8\text{cm}$ ,  $BC = 9\text{cm}$ ,  $AC = 10\text{cm}$ .

In  $\Delta AOB$ , X and Y are the mid-points of AO and BO.

$\therefore$  By using mid-point theorem, we have

$$XY = \frac{1}{2} AB = \frac{1}{2} \times 8\text{cm} = 4\text{cm}$$

Similarly, in  $\Delta BOC$ , Y and Z are the mid-points of BO and CO.

$\therefore$  By using mid-point theorem, we have

$$YZ = \frac{1}{2} BC = \frac{1}{2} \times 9\text{cm} = 4.5\text{cm}$$

And, in  $\Delta COA$ , Z and X are the mid-points of CO and AO.

$$\therefore ZX = \frac{1}{2}AC = \frac{1}{2} \times 10\text{cm} = 5\text{cm}$$

Hence, the lengths of the sides of  $\Delta XYZ$  are  $XY = 4\text{cm}$ ,  $YZ = 4.5\text{ cm}$  and  $ZX = 5\text{cm}$ .

**Ans: 4.** Since PQRS is a square.

$\therefore PQ = QR \dots (i) [\because \text{sides of a square are equal}]$

Also,  $BQ = CR \dots (ii) [\text{given}]$

Subtracting (ii) from (i), we obtain

$$PQ - BQ = QR - CR$$

$$\Rightarrow PB = QC \dots (iii)$$

In  $\Delta APB$  and  $\Delta BQC$

$$AP = BQ$$

[given  $\angle APB = \angle BQC = 90^\circ$ ](each angle of a square is  $90^\circ$ )

$$PB = QC \text{ (using (iii))}$$

So, by using SAS congruence axiom, we have

$$\Delta APB \cong \Delta BQC$$

$$\therefore AB = BC \text{ [c.p.c.t.]}$$

Now, in  $\Delta ABC$

$$AB = BC \text{ [proved above]}$$

$$\therefore \angle ACB = \angle BAC = x^\circ \text{ (say) } [\angle\text{s opp. to equal sides}]$$

$$\text{Also, } \angle B + \angle ACB + \angle BAC = 180^\circ$$

$$\Rightarrow 90^\circ + x + x = 180^\circ$$

$$\Rightarrow 2x^\circ = 90^\circ$$

$$x^\circ = 45^\circ$$

$$\text{Hence, } \angle BAC = 45^\circ$$

**Ans: 5.**

~

D

Since DP and CP are angle bisectors of  $\angle D$  and  $\angle C$  respectively.

$$\therefore \angle 1 = \angle 2 \text{ and } \angle 3 = \angle 4$$

Now,  $AB \parallel DC$  and CP is a transversal

$$\therefore \angle 5 = \angle 1 \text{ [alt. int. } \angle\text{s]}$$

$$\text{But } \angle 1 = \angle 2 \text{ [given]}$$

$$\therefore \angle 5 = \angle 2$$

Now, in  $\Delta BCP$ ,  $\angle 5 = \angle 2$

$$\Rightarrow BC = BP \dots (i) \text{ [sides opp. to equal } \angle\text{s of a } \Delta]$$

Again,  $AB \parallel DC$  and DP is a transversal.

$$\therefore \angle 6 = \angle 3 \text{ (alt. int. } \Delta\text{s)}$$

$$\text{But } \angle 4 = \angle 3 \text{ [given]}$$

$$\therefore \angle 6 = \angle 4$$

Now, in  $\Delta ADP$ ,  $\angle 6 = \angle 4$

$\Rightarrow DA = AP \dots$  (ii) (sides opp. to equal  $\angle$ s of a  $\Delta$ )

Also,  $BC = DA \dots$  (iii) (opp. sides of parallelogram)

From (i), (ii) and (iii), we have

$$BP = AP$$

Hence, P is the mid-point of side AB.

### Assertion and Reason Answers-

1. a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
2. c) Assertion is correct statement but reason is wrong statement.

### Case Study Answers-

1.

|       |     |                                                                                          |
|-------|-----|------------------------------------------------------------------------------------------|
| (i)   | (b) | 30cm                                                                                     |
| (ii)  | (c) | 135, 135                                                                                 |
| (iii) | (c) | Trapezium has 1 pair of parallel sides, and parallelogram has 2 pairs of parallel sides. |
| (iv)  | (d) | equal.                                                                                   |
| (v)   | (a) | $BF = FC$                                                                                |

2.

|       |     |                     |
|-------|-----|---------------------|
| (i)   | (b) | 8cm                 |
| (ii)  | (a) | $24\text{cm}^2$     |
| (iii) | (c) | $360^\circ$         |
| (iv)  | (c) | Triangles.          |
| (v)   | (c) | Equal to $90^\circ$ |